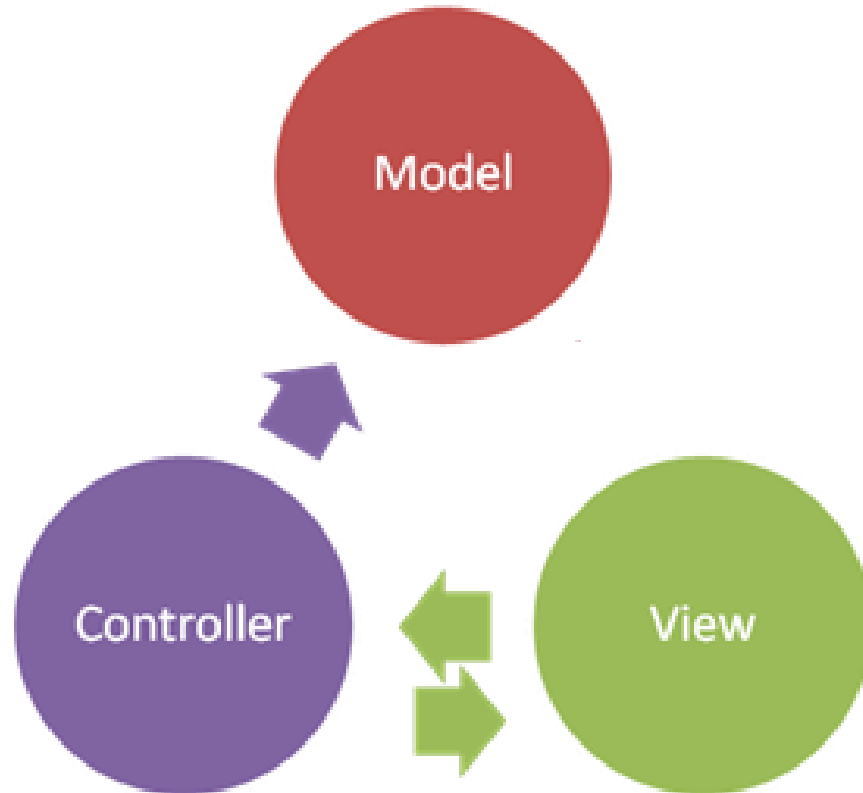
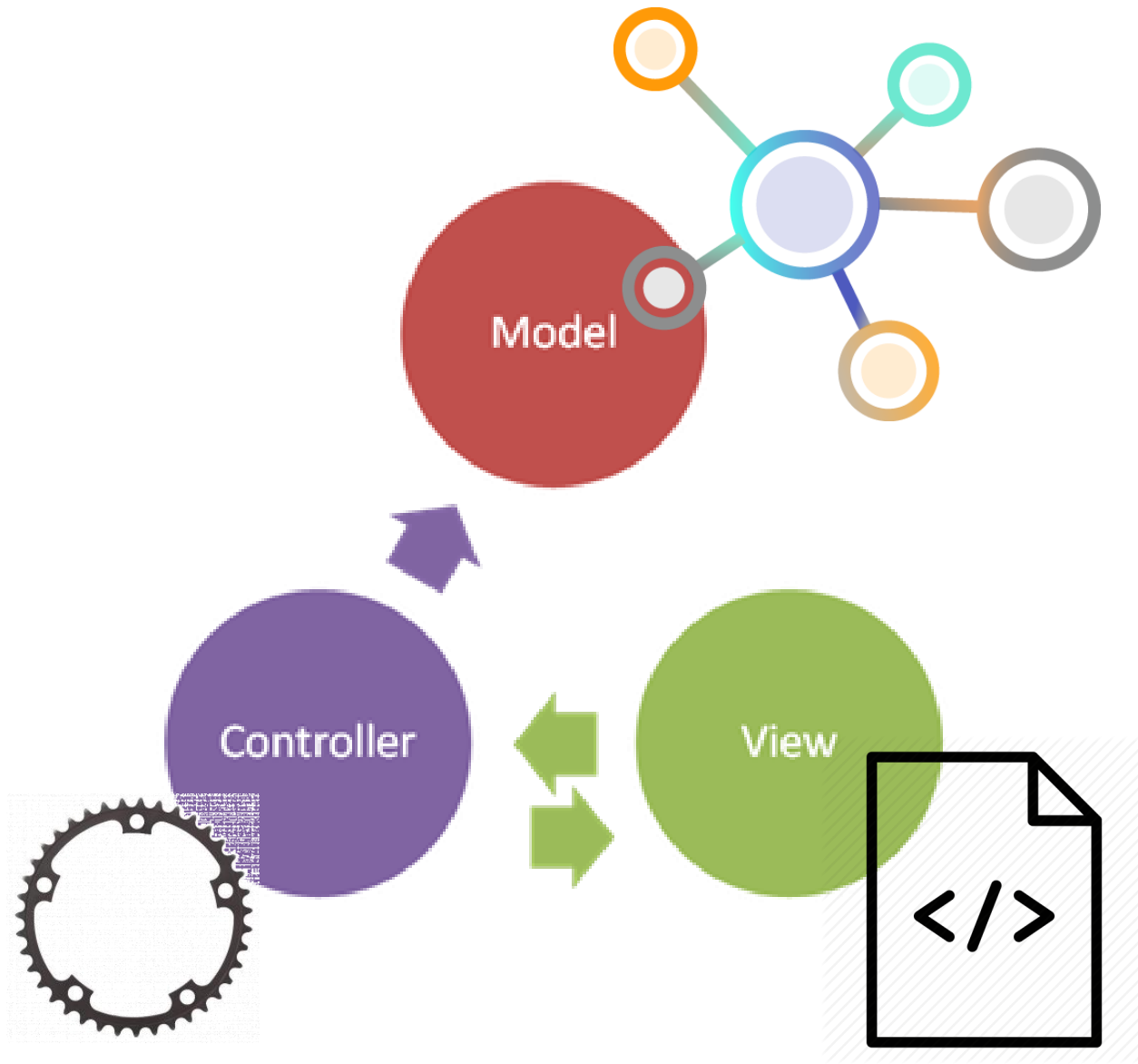


# MVC



# MVC



# Let's code a Todo-App with MVC in mind

The screenshot shows a web application interface for a todo list. It features a light blue header with the title 'Filter'. Below the title is a dropdown menu currently showing 'All Todos you have'. The main content area has a light blue background and contains a list of three todos: 'Master CSS', 'Learn HTML', and 'Become JS Ninja'. Each todo item is followed by a small square button with an 'x' icon for deletion. Below the list is a form with a text input field labeled 'Add Someting' and an 'Add' button. At the bottom, a grey bar displays the summary 'Total Todos: 3 Active: 2'.

**Filter**

All Todos you have ▾

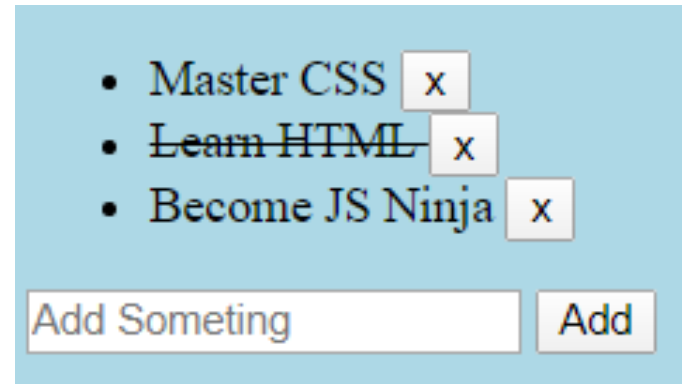
- Master CSS x
- Learn HTML x
- Become JS Ninja x

Add Someting Add

**Total Todos: 3 Active: 2**

# Initial plan

- onInit -> renderTodos
- onAddTodo
- onRemoveTodo
- onToggleTodo



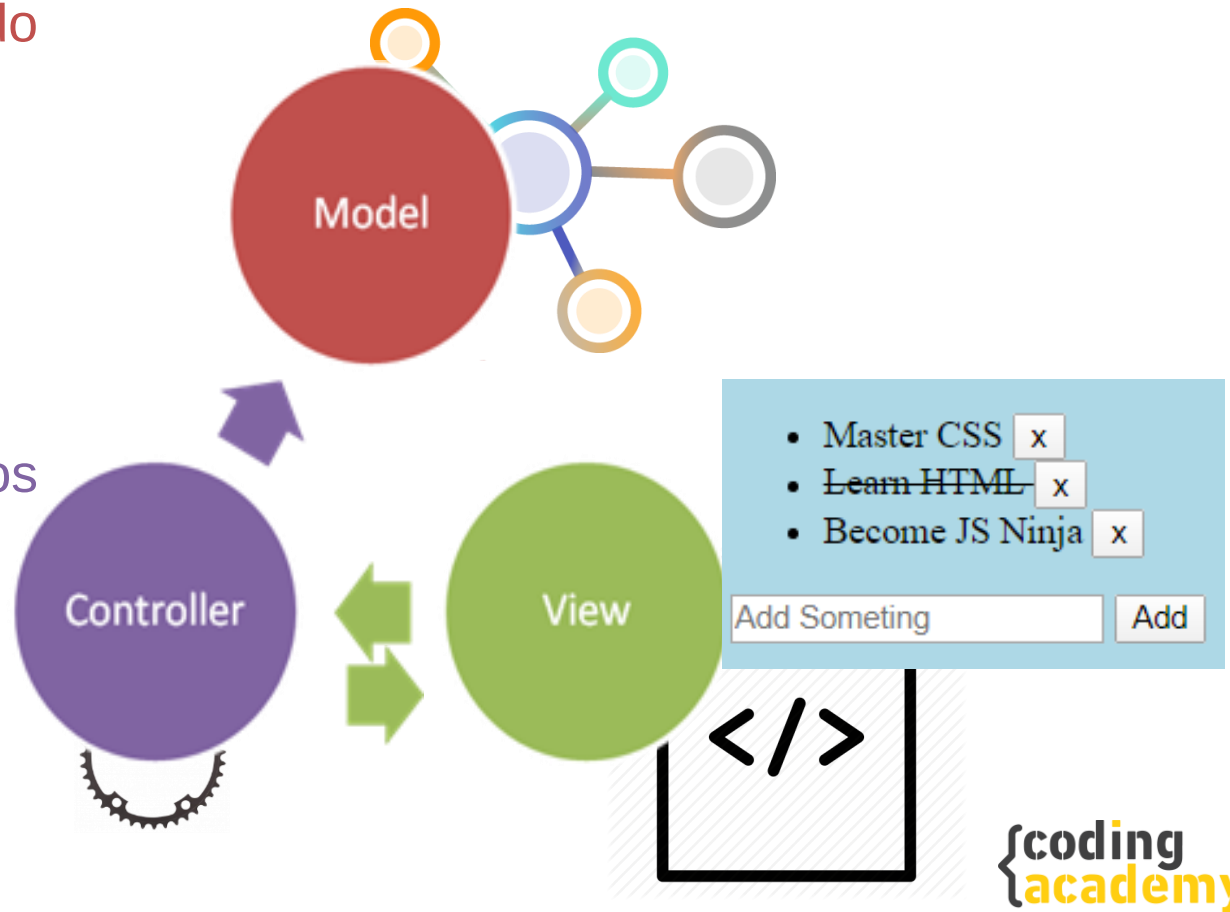
Let's do it

```
var todo = {  
  id: 't101',  
  txt: 'Fix that thing',  
  isDone: false,  
}
```

# Todo App Plan

- getTodosForDisplay
- addTodo
- removeTodo
- toggleTodo

- onInit -> renderTodos
- onAddTodo
- onRemoveTodo
- onToggleTodo



# <select>

```
<select onchange="onSetFilter(this.value)">  
  <option value="">All</option>  
  <option value="active">Active</option>  
  <option value="done">Done</option>  
</select>
```

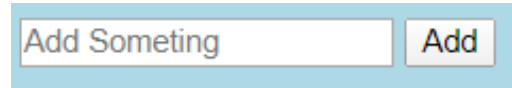
# event.stopPropagation()

Let's understand event propagation

```
<div class="box" oncontextmenu="alert('Box!')">  
  <div class="monster"  
    oncontextmenu="event.stopPropagation();  
                    event.preventDefault();  
                    alert('Aya');"  
  >  
</div>  
</div>
```

# event.preventDefault()

Some DOM events have default behavior, we sometimes want to prevent it



```
<form onSubmit="onAddTodo(event)">
  <input type="text" name="txt"
    placeholder="Add something" />
  <button>Add</button>
</form>
```

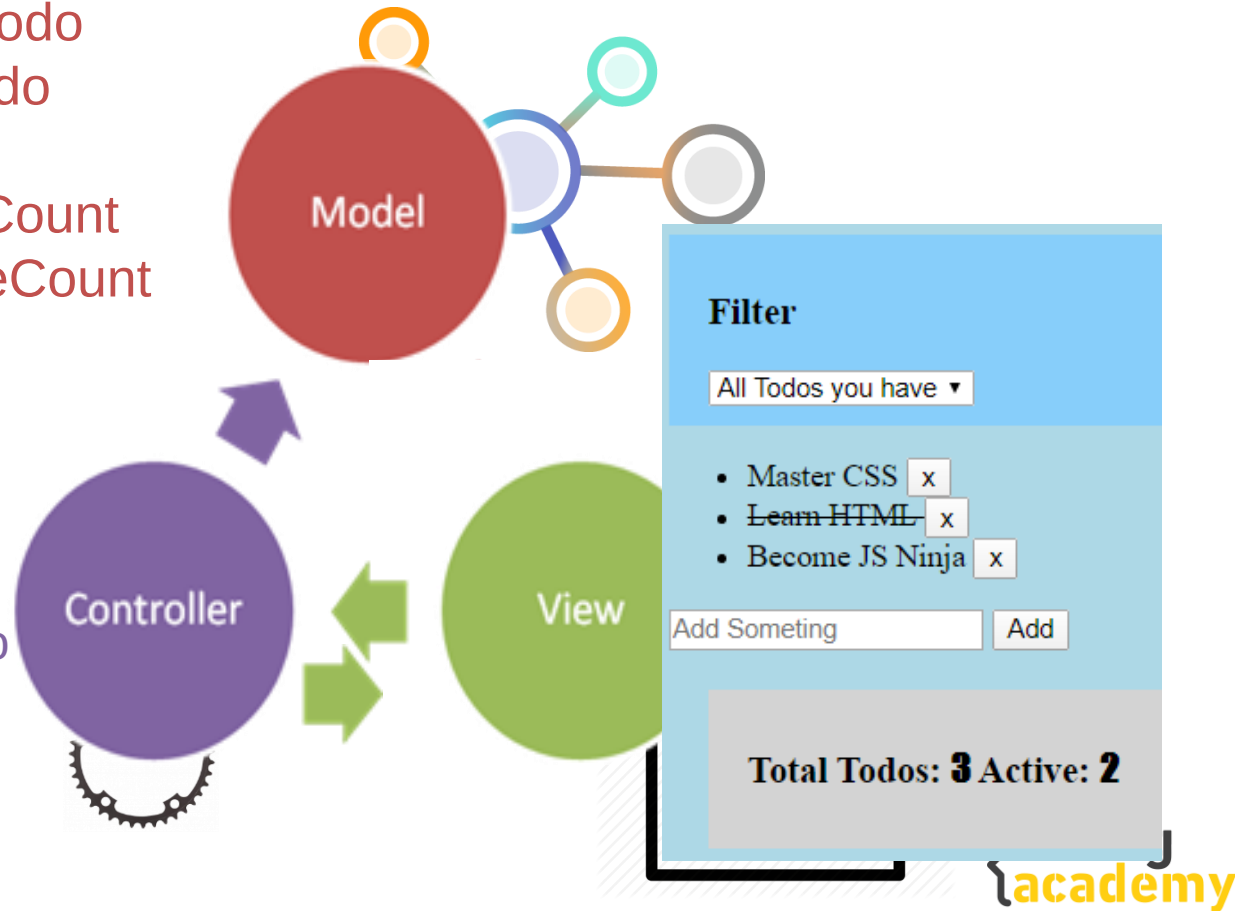
```
function onAddTodo(ev) {
  ev.preventDefault()
  const elTxt = document.querySelector('[name=txt]')
  const txt = elTxt.value
  console.log('Adding todo', txt)
  addTodo(txt)
  renderTodos()
  elTxt.value = ''
}
```



# Todo App - MVC

1. `_createTodos, _createTodo`
2. `getTodosForDisplay`
3. `addTodo`
4. `removeTodo`
5. `toggleTodo`
6. `setFilter`
7. `getTotalCount`
8. `getActiveCount`

1. `renderTodos`
2. `onInit`
3. `onAddTodo`
4. `onRemoveTodo`
5. `onToggleTodo`
6. `onSetFilter`

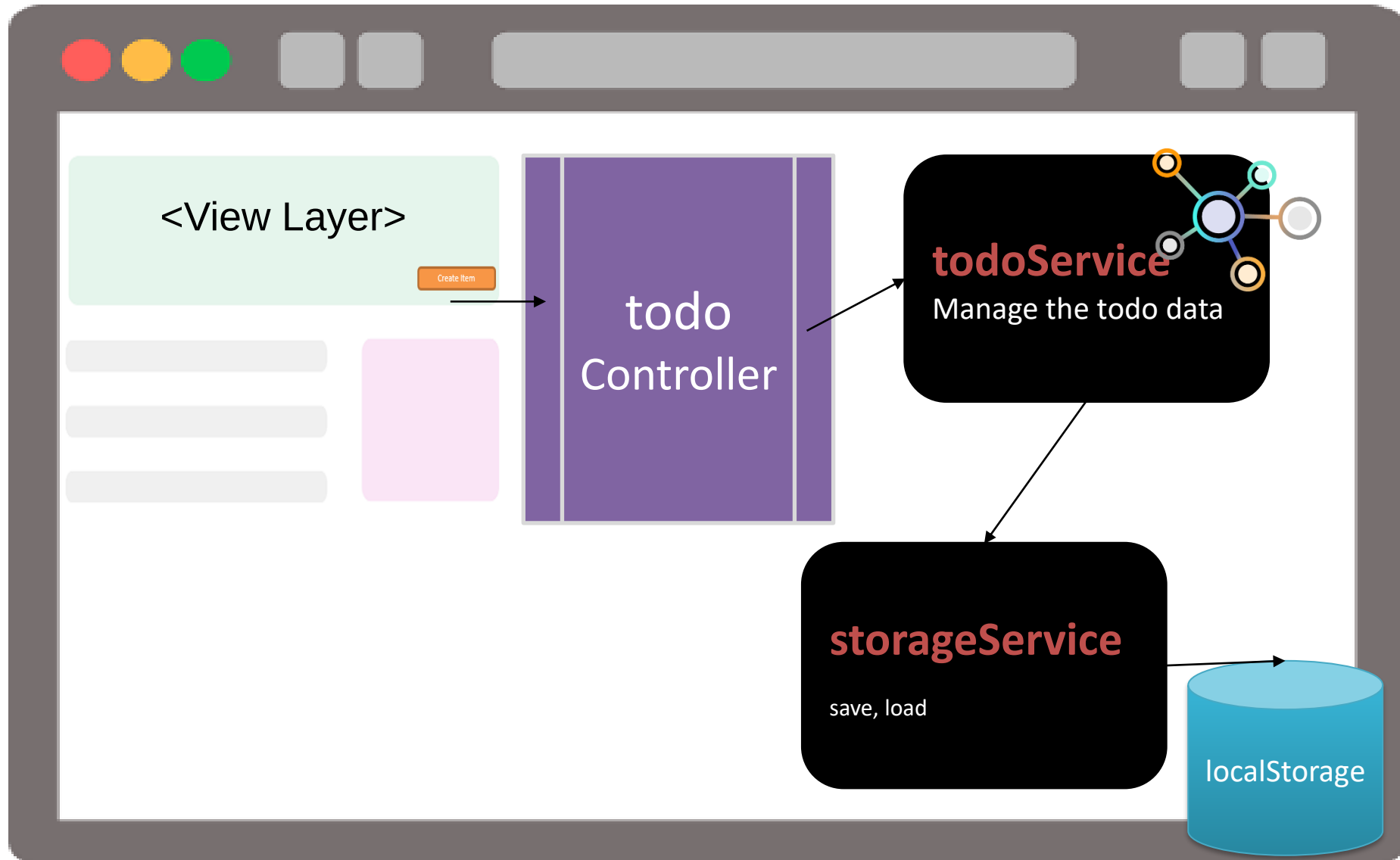


# Using localStorage

Lets build a storage-service:

```
function saveToStorage(key, value) {  
    var json = JSON.stringify(value)  
    localStorage.setItem(key, json)  
}  
  
function loadFromStorage(key) {  
    var json = localStorage.getItem(key)  
    var value = JSON.parse(json)  
    return value  
}
```

# Architecture block diagram



# MVC

